Minki Lee

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Integrated M.S. + Ph.D. Program Student
CompSec Lab [https://compsec.postech.ac.kr],
Department of Computer Science and Engineering,
Pohang University of Science and Technology (POSTECH), South Korea

Interests

- {Cyber-Physical Systems (CPS) | Distributed Systems | Embedded Systems} Security
- Secure-by-Design Development
- CPS Architecture & Engineering
- Open Source Software Security

Skills

- **Languages**: Korean (Native), English (Proficient)
- **Computer Languages**: C/C++, Python, JavaScript, PHP, Dart
- **Tools and Testbeds**: PX4/Pixhawk system, ROS2, gazebo, Crazyflie, Matlab simulations

Education

Integrated M.S. & Ph.D. Program — September 2025 - Present
Computer Security Lab, POSTECH CSE — *Pohang, South Korea*

- Advisor: Seulbae Kim
- Research in Cyber-Physical Systems' security
- Swarm system resilience and robustness

Bachelor of Engineering in Computer Science Major — February 2021 - August 2025
DGIST, School of Undergraduate Studies — *Daegu, South Korea*

- WAM/GPA: 3.60/4.30, percent scale: 93.0%
- Bachelor's Thesis-substitute Report: Security for Reliable Swarm Cyber-Physical System
- Courseworks: {Computer Security, Computer Network, Computer Architecture, Operating Systems, Computer Algorithms, Control Engineering, Real-time Embedded Systems}

Summer Session Visiting Student — June 2023 - August 2023
UC Berkeley, GPA 4.0 / 4.0 — *Berkeley, California*

- Visited by FGLP (Freshman Global Leadership Program), supported by DGIST

Completion of Technical Venture Leader Course — 23 February 2023
DGIST TVA (Technical Venture-leader Academy) — *Daegu, South Korea*

Publications

SwarmBox: A Plug-and-Play Drone Swarm Framework for Streamlined Development and Comprehensive Analysis (To Appear)
ACM International Conference on the Foundations of Software Engineering (FSE 2026), Montreal, Canada.
Minki Lee, Seojin Lee, Seulbae Kim

Demo: One Shot All Kill: Building Optimal Attack on Swarm Drones
Symposium on Vehicle Security and Privacy (VehicleSec 2024), San Diego, United States.
Minki Lee, Gangmin Kim, Jonghyun Kang, Hyunwoo Kim, Jangwon Lee, Hongjun Choi
https://dx.doi.org/10.14722/vehiclesec.2024.25008

Awards and Honors

Special Award: KIISC President's Award | *2025 ACDC (AI Cyber Defense Contest)* December 2025
- Awarded by the Korea Institute of Information Security & Cryptology (KIISC).
- Selected as a **Finalist** (Team RHCP) after ranking **5th** (student league) in the qualifier.
- *Focus:* Adversarial attacks and defense strategies for AI models (AI for Security).

Tech Award: KDIA President's Award | *HackTheDrone 2024* October 2024
- Awarded by the Korea Drone Innovation Association (Team CompSec).
- Ranked **7th** in Finals and **2nd** in the Qualifier.
- *Focus:* Drone protocol vulnerabilities, packet analysis, and RF signal jamming using SDR.

Grand Prize (DGIST President's Award) | *DGIST Startup Contest* October 2022
- Awarded for the most innovative business model among pre-startup teams.
- Won **2nd place** overall as a team leader.
- *Focus:* Proposed a Personalized Learning SaaS platform using AI-driven analytics.

Full National Scholarship | *DGIST* February 2021 – August 2025
- Awarded full tuition support for continuous academic excellence (8 semesters).

Experiences

[Research Experiences]

Visiting Undergraduate Intern June 2024 - August 2025
CompSec Lab, POSTECH *Pohang, South Korea*
- Advisor: Seulbae Kim
- Computer Security & CPS Security
- Research in Drone Swarm System Testbed

Undergraduate Student Researcher November 2022 – June 2024
Cyber-Physical Security (CPSec) Lab, DGIST *Daegu, South Korea*
- Advisor: Hongjun Choi
- Research about Secure and Robust Operation of Future Mobility (Drone Swarm)
- UGRP (Undergraduate Group Research Project) for Resilient Swarm Drones
- Presented Demo Poster at VehicleSec 2024, San Diego, United States

Research Internship December 2023 – January 2024
2024 Winter Internship Program, Department of EECS, DGIST *Daegu, South Korea*
- CSI Lab, Advisor: Kyung-Joon Park
- Research about Secure and Robust Future Mobility System

[Teaching Experiences]

Teaching Assistant February 2026 - June 2026
CSED415: Computer Security *POSTECH CSE*

Research (Teaching) Assistant August 2024 - December 2024
CSED199: Freshman Research Participation *POSTECH CSE*

Undergraduate Teaching Assistant August 2024 - December 2024
CSE401: Introduction to Database *School of Undergraduate Studies, DGIST*

[Work Experiences]

CUOP (Company - University Cooperation Program) Intern December 2024 – February 2025
Quester *Daegu, South Korea*
- Intern Developer
- Computer Vision AI model (object detection and tracking) for sensor fusion
- Experienced {YOLOv8, BiSeNet, SORT, DeepSORT, SAMURAI, SLAM}
- Product Security Analysis (Reverse Engineering)

CUOP (Company - University Cooperation Program) Intern June 2022 – August 2022
BostonAIMS *Seoul, South Korea*
- Intern Developer
- AI model Server Deployment / Development
- Huggingface, Dart / Flutter, Docker along with Firebase Server